



1. Identification of the substance/preparation and of the company undertaking

- **Trade Name:** KPL Di-Ammonium Phosphate (DAP)
- **Application:** High-analysis nitrogen-phosphorus fertilizer for agricultural use
- **Manufacturer/Supplier:**
Kesar Petroproducts Ltd.
Plot No. D-7/1, Lote Parshuram MIDC, Tal-Khed, Dist. Ratnagiri,
Maharashtra – 415722
- **Further Information:** Product Safety Department, Kesar Petroproducts Ltd.,
Ratnagiri Plant

2. Hazards Identification

- **GHS Classification:**
 - **Not classified** as flammable, explosive, or acutely toxic
 - **May cause eye/skin irritation** (H315, H319) due to alkaline nature
 - **Dust may cause respiratory irritation** (H335)
- **Label Elements:**
 - **Pictogram:** None required under standard fertilizer classification (GHS)
 - **Precautionary Statements:**
 - P261 – Avoid breathing dust
 - P280 – Wear protective gloves/eye protection
 - P305+P351+P338 – IF IN EYES: Rinse cautiously with water for several minutes

3. Composition/Information on Ingredients

- **Description:** Di-Ammonium Phosphate
- **Chemical Formula:** $(\text{NH}_4)_2\text{HPO}_4$
- **CAS No.:** 7783-28-0
- **Typical Purity (Fertilizer Grade):** ≥98%

- **Nutrient Content:**
 - **N (Nitrogen):** 18% minimum
 - **P₂O₅ (Phosphorus Pentoxide):** 46% minimum
 - **pH (1% solution):** 7.8 – 8.2 (slightly alkaline)
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4. First-Aid Measures

- **Inhalation:** Move to fresh air. If coughing or discomfort persists, seek medical attention.
 - **Skin Contact:** Wash thoroughly with soap and water. Remove contaminated clothing.
 - **Eye Contact:** Flush immediately with plenty of water for at least 15 minutes. Lift eyelids occasionally. Seek medical advice.
 - **Ingestion:** Rinse mouth. Do NOT induce vomiting. Give water to drink if conscious. Seek medical help.
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5. Fire-Fighting Measures

- **Extinguishing Media:** Water spray, dry chemical, CO₂
 - **Special Hazards:** May release ammonia and oxides of phosphorus when exposed to high heat
 - **Firefighter Protection:** Wear full protective gear and self-contained breathing apparatus
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6. Accidental Release Measures

- **Personal Precautions:** Wear dust mask, safety goggles, and gloves
 - **Environmental Precautions:** Avoid discharge into waterways or soil in large quantities (eutrophication risk)
 - **Cleanup Methods:** Sweep or vacuum (wet method preferred). Place in sealed container. Do not allow runoff to enter drains.
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7. Handling and Storage

- **Handling:** Minimize dust generation. Use in well-ventilated areas. Avoid contact with strong acids or bases.
 - **Storage:** Store in a cool, dry, well-ventilated area in sealed bags. Keep away from incompatible materials (e.g., lime, urea in bulk storage to prevent caking or ammonia loss).
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8. Exposure Controls/Personal Protection

- **Occupational Exposure:** No specific OEL; general industrial dust limits apply
- **Engineering Controls:** Local exhaust ventilation at bagging/dosing stations

- **PPE:**
 - Respiratory: Dust mask (N95) if airborne dust is visible
 - Hand Protection: Nitrile or rubber gloves
 - Eye Protection: Safety goggles
 - Skin Protection: Long sleeves, apron, or coveralls
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9. Physical and Chemical Properties

- **Appearance:** White to off-white granular solid
 - **Odour:** Ammoniacal (faint, especially when damp or heated)
 - **pH (1% aqueous solution):** 7.8 – 8.2
 - **Bulk Density:** ~0.9–1.0 g/cm³
 - **Solubility in Water:** Highly soluble (~58 g/100 mL at 25°C)
 - **Decomposition:** Releases ammonia above 70°C
 - **Hygroscopicity:** Moderately hygroscopic (may cake if exposed to moisture)
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10. Stability and Reactivity

- **Stable under normal conditions**
 - **Incompatible with:** Strong acids (releases ammonia gas), strong alkalis, calcium/magnesium fertilizers in solution
 - **Hazardous Decomposition:** Ammonia (NH₃), phosphorus oxides (at high temp)
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11. Toxicological Information

- **Acute Toxicity (oral, rat):** LD₅₀ > 2,000 mg/kg (low toxicity)
 - **Skin/Eye Irritation:** Mild irritant due to alkalinity
 - **Inhalation:** Dust may cause nasal/throat irritation
 - **Chronic Effects:** Not classified as carcinogenic, mutagenic, or reproductive toxicant
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12. Ecological Information

- **Environmental Impact:** High phosphate content may cause **eutrophication** in water bodies
 - **Biodegradability:** Inorganic—does not biodegrade, but nutrients are assimilated by plants
 - **Precaution:** Store and apply in accordance with agricultural best practices to prevent runoff
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13. Disposal Considerations

- **Product:** Can be used as intended (fertilizer) or disposed of as non-hazardous industrial waste
 - **Contaminated Packaging:** Triple-rinse if empty; recycle or dispose per local rules
 - **Note:** Large quantities should not be landfilled near water sources
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14. Transportation Information

- **UN Number:** Not regulated (fertilizer grade)
 - **Proper Shipping Name:** Ammonium phosphate, dibasic
 - **Transport Classification:** Not restricted under IMDG, IATA, or ADR for standard agricultural use
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15. Regulatory Information

- **GHS Status:** Not classified as hazardous under EU CLP or OSHA HCS for fertilizer-grade material
 - **Fertilizer Regulation:** Compliant with Fertilizer (Control) Order, India
 - **REACH/TSCA:** Listed
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16. Other Information

This SDS is based on current scientific understanding. It is not a warranty of safety or performance. Users must comply with national and local regulations for storage, handling, and application.

Issued by: Quality Control Department, Kesar Petroproducts Ltd.

Date: 29 December 2025